AgentGuard

Sprint Planner v1.0

AI Agent Governance Layer

"Think Palo Alto Prisma, but for AI agents."

Owner	Praveen | Stridemap.ai
Mode	Solopreneur · Full-Time · Vibe-Coding with Claude Code
Timeline	24 Sprints · ~24 Weeks · Full-time weekdays
Target	Demo-ready by Week 4 | v1.0 Release by Week 8
Confidentiality	CONFIDENTIAL — Not for external distribution
Version	1.0 | April 2026

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Strategic Overview

Three-Phase Roadmap

Phase	Sprints	Weeks	Theme	Key Milestone
MVP — Fast-Path Demo	1–4	1–4	Foundation → Interceptor → Policy/DLP → Injection/Audit	■ Demo video + OSS launch
Option A · Phase 1 Governance Layer	5–8	5–8	FastAPI + Postgres + SIEM + LLM Classifier	■ v1.0 — production REST API
Option A · Phase 2 Pilot Ready	9–12	9–12	SSO/RBAC + Kubernetes + Next.js Dashboard	■ v1.5 — first pilot customer
Option A · Phase 3 Production Scale	13–16	13–16	PyPI + Policy Marketplace + Multi-agent	■ v2.0-alpha — public OSS launch
Option C Runtime Security	17–24	17–24	OS sandboxing + eBPF + process isolation	■■ v2.0 — full runtime security

Solopreneur Daily Schedule

Time Block	Activity	Notes
9:00 – 10:30 AM	■ Learning Deep Dive	Read PRD/Architecture/Design for day's module; take interview notes
10:30 AM – 1:00 PM	■ Claude Code Session 1	Core implementation — paste spec into context, vibe-code
1:00 – 1:45 PM	■ Lunch + Async	LinkedIn, emails, community engagement
1:45 – 4:30 PM	■ Claude Code Session 2	Tests, integration, and polish
4:30 – 5:00 PM	■ Sprint Journal	Commit all code, update Linear tickets, write 3 interview talking points

Vibe-Coding Strategy with Claude Code

- Start each module by pasting the relevant PRD requirements + functional spec into Claude Code context.
- Use CLAUDE.md at repo root to maintain architecture context across all sessions.
- Commit frequently — each Claude Code session ends with a green CI pipeline.
- Don't skip tests — they are your interview differentiator AND your quality gate.
- The 9–10:30 AM Learning Deep Dive is protected time — this is your interview prep compound interest.

Key Decisions & Tradeoffs

Decision	Chosen	Rejected	Rationale
Storage MVP	SQLite	PostgreSQL from day 1	Zero-setup DX; abstraction allows upgrade

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Decision	Chosen	Rejected	Rationale
Dashboard MVP	Streamlit	Next.js from day 1	Ship demo in Week 2; Next.js in Sprint 11
LLM Classifier	Ollama local (dev)	Cloud-only from day 1	$0 dev cost; provider swap is one env var
Injection detection	Two-stage (regex + LLM)	LLM-only	Regex handles 95% of patterns at <5ms
Data handling	VPC-only, zero telemetry	Cloud callbacks	Enterprise non-negotiable
Sprint ordering	Policy then Injection (S3 then S4)	Parallel dev	Policy engine is prerequisite for injection action

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PHASE 1 — MVP: FAST PATH TO DEMO (Sprints 1–4 · Weeks 1–4)

Goal: Working demo of all 5 core modules by end of Week 4. Demo video on LinkedIn. OSS launch on GitHub. Job search signal begins.

SPRINT 1 — WEEK 1 — MVP

Foundation & Landing Page

Sprint Goal	GitHub monorepo live · stridemap.ai deployed on Vercel · CI pipeline green · full folder scaffold
Demo Moment	stridemap.ai publicly accessible — your first market signal
Phase	MVP
Week	1

■ LEARNING DEEP DIVE (Monday 9–10:30 AM)

- PRD Sections 1–4: Executive Summary, Vision, Personas, Architecture Overview
- Roadmap Sections 1–5: Executive Summary, Competitive Position, Tech Stack, Monorepo Architecture, Sprint 1 daily plan
- Key concept: Why tool-call layer > model layer for governance; NGFW mental model applied to AI agents; fail-open design philosophy
- Write 3 interview talking points about the problem space in your sprint journal

■ CLAUDE CODE VIBE-CODING SESSIONS

Session 1 (Mon): Repo scaffold

```
Create GitHub monorepo 'agentguard'. Directories: web/ (Next.js 14), agentguard/ (Python package), dashboard/ (Streamlit), demo/, policies/, tests/, docs/, .github/workflows/. Include .gitignore, .env.example. pyproject.toml for agentguard package (Python 3.11, MIT license). First commit and push.
```

Session 2 (Tue-Wed): Next.js landing page + Vercel deploy

```
Professional landing page for stridemap.ai — 'The Palo Alto Prisma for AI Agents'. Sections: Hero, Problem (4 cards), Solution (5-module diagram), How It Works (3 steps), Competitor table, Early access form. Tailwind CSS, dark theme #00D4AA accent. Deploy to Vercel, connect custom domain.
```

Session 3 (Thu): GitHub Actions CI/CD

```
python-ci.yml: Python 3.11, ruff + black, pytest --cov=agentguard --cov-fail-under=80. web-deploy.yml: Node 18, ESLint, TypeScript check, next build. Both trigger on path-scoped pushes.
```

Session 4 (Fri): README + scaffold

```
Root README.md with CI badge, coverage badge, elevator pitch, 30-second quickstart, Mermaid architecture diagram, competitor table, contributing guide. Scaffold all Python folder stubs with README files.
```

■ LINEAR TICKETS

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-001	Create GitHub monorepo with full folder scaffold	P0	module:infra, type:feature	2	Mon

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-002	Add Next.js 14 landing page to web/ directory	P0	module:dashboard, type:feature	3	Tue
AGD-003	Deploy web/ to Vercel and connect stridemap.ai domain	P0	module:infra, type:feature	2	Wed
AGD-004	Professional email via Zoho Mail (hello@ + security@)	P1	module:infra, type:feature	1	Wed
AGD-005	GitHub Actions CI pipeline for Python (lint + test + cov)	P0	module:infra, type:feature	2	Thu
AGD-006	GitHub Actions CI pipeline for web/ (ESLint + TS + build)	P1	module:infra, type:feature	1	Thu
AGD-007	Write comprehensive root README.md with badges	P1	type:docs	2	Fri
AGD-008	Scaffold Python folders with stub README + conftest.py	P1	type:docs, module:infra	1	Fri
AGD-009	Update LinkedIn + post "stridemap.ai is live"	P2	type:demo	1	Fri

■ ACCEPTANCE CRITERIA / SPRINT GATE

- ■ github.com/stridemap/agentguard is public and accessible
- ■ stridemap.ai loads with HTTPS on desktop and mobile
- ■ hello@stridemap.ai and security@stridemap.ai emails work
- ■ GitHub Actions CI runs green on every push to main
- ■ All folder stubs exist with placeholder README files
- ■ Zero secrets or API keys committed to the repo (CI check)

■ INTERVIEW TALKING POINTS

1. Monorepo strategy: One repo for Python library, landing page, dashboard, demos and tests — when a hiring manager clicks the GitHub link they see a complete product, not scattered repos.

2. CI from day one: Enforced 80% test coverage as branch-protection gate from Sprint 1, before writing a single line of product code — mirrors how Palo Alto tests firewall policy changes before production.

3. Cost discipline: Entire foundation sprint cost $0 using Vercel free, GitHub free, Zoho Mail free — same 'zero-cost to start' principle applied to infrastructure as to the open-source library.

SPRINT 2 — WEEK 2 — MVP

LangGraph Interceptor + Event Store + Trace Dashboard

Sprint Goal	Working LangGraph interceptor logging every tool call to SQLite · Live Streamlit trace dashboard auto-refreshing every 2 seconds

Demo Moment	■ Screen recording: 3-agent system running, Streamlit dashboard shows every tool call in real time with agent ID, tool name, latency, and risk flags
Phase	MVP
Week	2

■ LEARNING DEEP DIVE (Monday 9–10:30 AM)

- PRD Section 5 (Module 1: Agent Monitor & Trace Viewer) — all 9 requirements AM-01 through AM-09
- System Architecture Section 5.1 (Interceptor Layer) + 5.5 (Event Store & Async Queue)
- Functional Spec: UC-01 through UC-08 (use case table)
- Study: LangGraph BaseCallbackHandler source code (GitHub: langchain-ai/langgraph)
- Key concept: Why intercept at on_tool_start/on_tool_end and NOT at the LLM layer

■ CLAUDE CODE VIBE-CODING SESSIONS

Session 1 (Mon-Tue): Core interceptor

```
agentguard/interceptor.py subclassing BaseCallbackHandler. Implement on_tool_start + on_tool_end. Fail-open: ALL exceptions caught → WARNING log → tool call NEVER blocked. Async asyncio.Queue for non-blocking writes (<5ms p99). Pydantic v2 AgentEvent schema: event_id (UUID4), timestamp (ISO8601), agent_id, session_id, tool_name, input_hash (SHA-256), output_hash (SHA-256), latency_ms, risk_flags[], policy_result.
```

Session 2 (Tue-Wed): SQLite event store

```
agentguard/event_store.py. Append-only SQLite (no UPDATE/DELETE). Background thread consumer. health_check() → {status, db_connection, queue_depth, events_per_second, version}. Risk auto-flagging: external_domain, large_payload (>10KB), sensitive_keyword, high_frequency (>10/min), unknown_tool.
```

Session 3 (Wed-Thu): 3-agent finance demo

```
demo/finance_scenario.py. 3 agents: EmailReaderAgent → CRMLookupAgent → ResponseWriterAgent. AgentGuard injected via config={'callbacks': [AgentGuardInterceptor()]}. ZERO changes to agent logic. Realistic PII in mock tool outputs (SSN, email, phone).
```

Session 4 (Thu-Fri): Streamlit dashboard v1

```
dashboard/app.py. Auto-refresh every 2 seconds. Live event feed table (last 100 events) colour-coded: green=CLEAN, yellow=SUSPICIOUS, red=BLOCKED. Filters: agent_id, tool_name, risk_flag, time range. Event detail panel on click. Metrics: total events, p99 latency, events/sec, high-risk count.
```

■ LINEAR TICKETS

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-010	Implement BaseCallbackHandler interceptor (on_tool_start/end)	P0	module:interceptor, type:feature	5	Mon-Tue
AGD-011	Fail-open guarantee — exceptions never block tool call	P0	module:interceptor, type:feature	3	Mon
AGD-012	Structured event logging with Pydantic v2 event schema	P0	module:interceptor, type:feature	3	Tue
AGD-013	Async non-blocking write queue (asyncio.Queue, <5ms p99)	P0	module:interceptor, type:feature	3	Tue
AGD-014	SQLite event store with append-only design	P0	module:interceptor, type:feature	3	Wed
AGD-015	Risk auto-flagging (5 risk types)	P1	module:interceptor, type:feature	3	Wed

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ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-016	3-agent finance demo scenario	P1	module:interceptor, type:feature	3	Wed-Thu
AGD-017	Streamlit trace dashboard v1 (auto-refresh, filters, colouring)	P1	module:dashboard, type:feature	5	Thu-Fri
AGD-018	Unit tests: test_interceptor.py + test_event_store.py	P0	module:interceptor, type:test	3	Fri
AGD-019	Health endpoint: GET /health returns JSON status	P1	module:interceptor, type:feature	1	Fri
AGD-020	Record 2-min Loom demo + post to LinkedIn	P1	type:demo	1	Fri

■ ACCEPTANCE CRITERIA / SPRINT GATE

- ■ 100 consecutive tool calls → 100 events in DB (zero dropped)
- ■ Interceptor forced to throw RuntimeError → tool call completes normally, WARNING event logged
- ■ Latency overhead over 1,000 calls: p50 < 5ms, p95 < 20ms, p99 < 50ms
- ■ Streamlit dashboard loads in <3 seconds with 10,000 events
- ■ Risk flag auto-detection identifies all 5 risk types in scenario tests
- ■ Queue depth stable (no growth) under sustained 100 events/sec load
- ■ pytest --cov=agentguard --cov-fail-under=80 passes

■ INTERVIEW TALKING POINTS

1. Callback architecture: AgentGuard uses LangGraph's BaseCallbackHandler — same hook system LangSmith uses for observability, but I go further: evaluate every tool call against security policies and write to an append-only cryptographic audit chain.

2. Fail-open design: The most important architectural decision was fail-open. If AgentGuard crashes, agents keep running. Modeled on Palo Alto NGFW bypass mode — security infrastructure must never become a single point of failure.

3. Async non-blocking logging: Tool calls can't wait for database I/O. asyncio.Queue as in-memory buffer — interceptor enqueues in <5ms, background thread flushes to SQLite. Same pattern as high-performance network logging systems (Cisco background).

SPRINT 3 — WEEK 3 — MVP

Policy Engine + DLP Scanner

Sprint Goal	YAML policy rules evaluated against every tool call · Microsoft Presidio DLP scanning inputs and outputs · ALLOW/BLOCK/ALERT/BYPASS decisions enforced
Demo Moment	■ Agent tries to email a document containing SSN → DLP catches it → policy engine returns BLOCK → tool call denied → audit event written → dashboard shows red BLOCKED event
Phase	MVP
Week	3

■ LEARNING DEEP DIVE (Monday 9–10:30 AM)

- PRD Section 6 (Module 2: Policy Engine & DLP) — all requirements PE-01 through PE-07, DLP-01 through DLP-06
- System Architecture Section 5.2 (Policy Engine) + 5.3 (DLP Scanner — Microsoft Presidio)
- Study: Microsoft Presidio documentation (presidio-analyzer, presidio-anonymizer)
- Study: PyYAML + Pydantic v2 for YAML schema validation
- Key concept: Policy-as-Code pattern and how it mirrors ACL rules in NGFW (direct parallel to Cisco/Palo Alto background)

■ CLAUDE CODE VIBE-CODING SESSIONS

Session 1 (Mon-Tue): Policy Engine

```
agentguard/policy_engine.py. PolicyEngine class: load_policies(path), hot_reload() via watchdog, evaluate(event) -> PolicyDecision. In-memory only, <5ms p99. YAML schema: id, name, priority, conditions (tool_name regex, destination_domain, data_pattern, direction, agent_id), action (ALLOW/BLOCK/ALERT/BYPASS), alert_severity. Sample policies/default.yaml with 10 enterprise rules.
```

Session 2 (Tue-Wed): DLP Scanner

```
agentguard/dlp_scanner.py using Microsoft Presidio. DLPScanner.scan(text, direction) -> DLPResult(entities_found, redacted_text, risk_level, latency_ms). 15+ PII entities: PERSON, EMAIL, PHONE, SSN, CREDIT_CARD, MEDICAL_LICENSE, IP_ADDRESS, etc. 100% local — zero external API calls. <50ms for 10KB payload. Sensitivity levels: LOW/MEDIUM/HIGH.
```

Session 3 (Wed-Thu): Integration + healthcare demo

```
Wire PolicyEngine + DLPScanner into interceptor pipeline. on_tool_start: DLP scan input, PolicyEngine evaluate, BLOCK raises exception caught by fail-open wrapper. on_tool_end: DLP scan output, log redacted version if PII found. Healthcare demo: PHI exfiltration attempt → BLOCKED.
```

Session 4 (Thu-Fri): Tests + policies

```
test_policy_engine.py: 10 scenario tests, hot-reload (new rule in <2s), invalid YAML error, priority ordering, performance (1000 evals <5s). test_dlp_scanner.py: all 15 entity types, FP rate <2%, redaction output. Create policies/finance.yaml + policies/healthcare.yaml.
```

■ LINEAR TICKETS

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-021	Policy Engine: YAML loader with Pydantic v2 validation	P0	module:policy, type:feature	5	Mon
AGD-022	Policy Engine: in-memory rule evaluation (<5ms p99)	P0	module:policy, type:feature	3	Tue
AGD-023	Policy Engine: ALLOW/BLOCK/ALERT/BYPASS decision engine	P0	module:policy, type:feature	3	Tue
AGD-024	Policy Engine: hot-reload without agent restart	P1	module:policy, type:feature	2	Wed
AGD-025	DLP Scanner: Microsoft Presidio integration (100% local)	P0	module:dlp, type:feature	5	Tue-Wed
AGD-026	DLP Scanner: 15+ PII entity type detection	P0	module:dlp, type:feature	3	Wed
AGD-027	DLP Scanner: redaction mode (REDACT/ALERT/BLOCK)	P0	module:dlp, type:feature	2	Wed
AGD-028	Wire PolicyEngine + DLPScanner into interceptor pipeline	P0	module:policy, type:feature	3	Thu

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ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-029	Sample YAML: default.yaml + finance.yaml + healthcare.yaml	P1	module:policy, type:docs	2	Thu
AGD-030	Healthcare demo (PHI exfiltration → BLOCKED)	P1	module:policy, type:feature	3	Thu
AGD-031	Unit tests: test_policy_engine.py (10 scenarios + perf)	P0	module:policy, type:test	3	Fri
AGD-032	Unit tests: test_dlp_scanner.py (entity types + FP rate)	P0	module:dlp, type:test	3	Fri

■ ACCEPTANCE CRITERIA / SPRINT GATE

- ■ All 10 default policy rules evaluate correctly in unit tests
- ■ Policy hot-reload: updated rule applied within 2 seconds without restart
- ■ DLP detects SSN, CREDIT_CARD, EMAIL, PHONE, MEDICAL_LICENSE in test corpus
- ■ DLP false positive rate < 2% on synthetic non-PII corpus (50 samples)
- ■ Policy evaluation: 1,000 evaluations complete in < 5 seconds (in-memory)
- ■ DLP scan: < 50ms for 10KB payload
- ■ Healthcare scenario demo: PHI exfiltration attempt → BLOCKED → logged

■ INTERVIEW TALKING POINTS

1. Policy-as-Code: Built a YAML policy format any security engineer who has written ACL rules can understand in 5 minutes. Policy files are version-controlled alongside agent code — directly from 'Policy as Code' principle from network security work.

2. Local DLP is non-negotiable: Every enterprise CTO's first question is 'does our patient data leave our network?' AgentGuard answers: never. Microsoft Presidio runs 100% inside the customer VPC — chosen over cloud-based DLP APIs (AWS Comprehend, Google DLP) for this reason.

3. Four-action decision model: ALLOW/BLOCK/ALERT/BYPASS maps exactly to how NGFW handles traffic. ALERT proceeds but notifies; BYPASS skips AgentGuard entirely for trusted internal tools. Security engineers understand this immediately.

SPRINT 4 — WEEK 4 — MVP

Prompt Injection Detector + Audit Chain + Demo Polish + OSS Launch

Sprint Goal	Two-stage injection detection · SHA-256 cryptographic audit chain · Full 3-scenario attack demo · PyPI package stub published
Demo Moment	■ THE BIG DEMO: 3 named attack scenarios (PII_EXFIL, INJECT, UNAUTH) shown live with dashboard blocking/logging each attack. 5-min Loom video on LinkedIn.
Phase	MVP
Week	4

■ LEARNING DEEP DIVE (Monday 9–10:30 AM)

- PRD Section 7 (Module 3: Prompt Injection) + Section 8 (Module 4: Audit Chain)
- System Architecture Section 5.4 (Injection Detector) + 5.6 (Audit Chain)
- Study: OWASP LLM Top 10 (LLM01–LLM06) — critical for interview conversations
- Study: SHA-256 cryptographic chaining concept (like blockchain but without consensus overhead)
- Key concept: Two-stage detection rationale — fast regex first, expensive LLM only when Stage 1 = SUSPICIOUS

■ CLAUDE CODE VIBE-CODING SESSIONS

Session 1 (Mon-Tue): Two-stage injection detector

```
agentguard/injection_detector.py. Stage 1 regex library (<5ms p99): direct injection (ignore instructions, DAN, role-play), indirect (HTML comments, base64, JSON injection), exfiltration patterns. Returns CLEAN/SUSPICIOUS/INJECTED. Stage 2 LLM classifier via LLMClient (only fires when Stage 1=SUSPICIOUS). Sensitivity: LOW (Stage 1 only), MEDIUM, HIGH (always both).
```

Session 2 (Tue): LLM Client abstraction

```
agentguard/llm_client.py. Abstract LLMClient base class. OllamaClient (llama3.2), OpenAIClient (gpt-4o-mini), AnthropicClient (claude-haiku). Provider via AGENTGUARD_LLM_PROVIDER env var (default: ollama). 10s timeout, 3-attempt exponential backoff, graceful degradation if unavailable.
```

Session 3 (Wed): SHA-256 audit chain

```
Upgrade event_store.py: for each event: prev_hash = stored hash of previous event. event_hash = SHA-256(json.dumps(event_content, sort_keys=True) + prev_hash). verify_chain() iterates all events, recomputes expected hash vs stored. Returns ChainVerificationResult(is_valid, total_events, first_tampered_id). CLI: python -m agentguard verify-chain.
```

Session 4 (Thu-Fri): 3 attack scenarios + PyPI

```
PII_EXFIL: financial agent emails SSN/CC to external address → DLP catches → BLOCK. INJECT: legal agent reads contract with embedded 'Ignore previous instructions...' → INJECTED → quarantine. UNAUTH: IT agent calls unauthorized API → policy BLOCK. Coloured terminal output. pyproject.toml for PyPI publication.
```

■ LINEAR TICKETS

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-033	Injection Detector: Stage 1 regex library (OWASP LLM01-06)	P0	module:injection, type:feature	5	Mon
AGD-034	Injection Detector: Stage 2 LLM classifier	P0	module:injection, type:feature	3	Mon-Tue
AGD-035	LLM Client abstraction (Ollama/OpenAI/Anthropic via env var)	P0	module:injection, type:feature	3	Tue
AGD-036	Wire Injection Detector into interceptor pipeline	P0	module:injection, type:feature	2	Tue
AGD-037	SHA-256 cryptographic audit chain (full, not stub)	P0	module:audit, type:feature	5	Wed
AGD-038	verify_chain() CLI command + ChainVerificationResult model	P0	module:audit, type:feature	3	Wed
AGD-039	Demo: PII_EXFIL attack scenario	P0	module:interceptor, type:feature	3	Thu
AGD-040	Demo: INJECT attack scenario (contract injection)	P0	module:injection, type:feature	3	Thu

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ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-041	Demo: UNAUTH attack scenario (unauthorized API)	P0	module:policy, type:feature	2	Thu
AGD-042	Unit tests: test_injection_detector.py (precision >0.90)	P0	module:injection, type:test	3	Fri
AGD-043	Unit tests: test_audit_chain.py (tamper detection)	P0	module:audit, type:test	3	Fri
AGD-044	PyPI package: pyproject.toml v0.1.0-alpha	P1	module:infra, type:feature	2	Fri
AGD-045	5-min Loom demo (3 attacks) + LinkedIn post	P0	type:demo	2	Fri

■ ACCEPTANCE CRITERIA / SPRINT GATE

- ■ Injection precision > 0.90, recall > 0.85 on 100-sample labelled corpus
- ■ Stage 1 regex scan: < 5ms p99 for 1KB input
- ■ Audit chain: verify_chain() detects any single-byte modification to any event
- ■ All 3 demo scenarios run without errors end-to-end
- ■ Dashboard shows all 3 attacks with correct risk colouring
- ■ pip install agentguard from local package succeeds
- ■ OWASP LLM01–LLM06 coverage > 80% in scenario tests
- ■ GitHub repo has >50 commits, CI green, coverage badge visible

■ INTERVIEW TALKING POINTS

1. Two-stage detection architecture: Regex handles 95% of known patterns at <5ms. LLM classifier only fires for edge cases when Stage 1 = SUSPICIOUS — keeps latency <50ms p99 while maintaining high recall. Same architecture as anti-spam systems.

2. Cryptographic audit chain: SHA-256 where each event's hash includes the previous — like a blockchain without consensus overhead. Any modification to any historical event breaks the chain at that point. Enterprises get tamper-evident logs for SOC2 and HIPAA.

3. OWASP LLM Top 10: Mapped every test scenario to an OWASP LLM category in ATTACK_MATRIX.md. When a CISO asks 'does this cover LLM01 through LLM06?' I can point to a documented coverage map — something most AI security tools cannot do.

■ DEMO GATE — End of Week 4 | Start Job Applications This Week

- ■ Record 5-min Loom demo video showing all 3 attack scenarios
- ■ Post to LinkedIn with #AIAgents #LangGraph #AISecurity #OpenSource
- ■ Submit agentguard to Hacker News 'Show HN'
- ■ DM 10 targeted enterprise security/AI professionals on LinkedIn with demo link
- ■ Update resume with AgentGuard project — tech stack and measurable results
- ■ Begin job applications: target 3 applications/week starting Week 5

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PHASE 2 — OPTION A: GOVERNANCE LAYER (Sprints 5–8 · Weeks 5–8)

Sprint 5 — FastAPI REST Layer + JWT Authentication

Week / Phase	5 · Option A Phase 1
Sprint Goal	Production-grade REST API wrapping AgentGuard core · JWT Bearer auth · 10 documented endpoints
Demo Moment	Postman collection showing all 10 endpoints with JWT auth — shareable with enterprise prospects

Key tickets (Sprint 5):

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-047	FastAPI app skeleton + Uvicorn server	P0	module:api, type:feature	3	Mon
AGD-048	JWT authentication (python-jose, 30-min expiry, roles)	P0	module:api, type:feature	5	Tue
AGD-049	10 REST API endpoints (events, audit, policies, health, metrics)	P0	module:api, type:feature	8	Wed-Thu
AGD-050	Prometheus /metrics endpoint (Prometheus-compatible)	P1	module:api, type:feature	3	Thu
AGD-051	RBAC enforcement (admin/analyst/auditor roles)	P0	module:api, type:feature	3	Fri
AGD-052	API integration tests (pytest + httpx TestClient)	P0	module:api, type:test	5	Fri

Sprint 6 — PostgreSQL Migration + Redis Async Queue

Week / Phase	6 · Option A Phase 1
Sprint Goal	Migrate event store from SQLite to PostgreSQL 15 · Redis queue for production throughput · Alembic migrations
Demo Moment	Locust load test showing 100+ events/sec sustained with zero queue growth

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-055	PostgreSQL 15 event store (asyncpg connection pool)	P0	module:infra, type:feature	5	Mon-Tue
AGD-056	Alembic migrations: initial schema + performance indexes	P0	module:infra, type:feature	3	Tue
AGD-057	Redis queue replacement for asyncio.Queue	P1	module:infra, type:feature	3	Wed
AGD-058	Docker Compose: postgres + redis + agentguard + dashboard	P0	module:infra, type:feature	3	Thu

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-059	Locust: 100 events/sec sustained test (PERF_BENCHMARKS.md)	P0	module:infra, type:test	3	Fri

Sprint 7 — SIEM/CEF Integration + Prometheus + Alerting

Week / Phase	7 · Option A Phase 1
Sprint Goal	High-severity events forwarded to SIEM as CEF webhooks · Prometheus metrics consumed by Grafana · Alerting hooks
Demo Moment	Splunk/Datadog webhook receiving real-time CEF alerts from AgentGuard demo scenario

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-062	SIEM webhook client (CEF format, configurable endpoint)	P0	module:api, type:feature	5	Mon-Tue
AGD-063	Splunk HEC + Datadog + Sentinel format adapters	P1	module:api, type:feature	3	Tue-Wed
AGD-064	Prometheus metrics (5 core metrics with labels)	P0	module:api, type:feature	3	Wed
AGD-065	Grafana dashboard JSON template	P1	module:api, type:docs	2	Thu
AGD-067	VPC-only egress test: zero outbound when SIEM not configured	P0	module:api, type:test	2	Fri

Sprint 8 — LLM Classifier Upgrade + v1.0.0 PyPI Release

Week / Phase	8 · Option A Phase 1
Sprint Goal	Injection eval corpus + precision/recall gates · Policy v2 regex conditions · v1.0.0 published on PyPI
Demo Moment	■ pip install agentguard==1.0.0 on clean environment succeeds

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-068	Injection corpus: 100-sample labelled dataset	P0	module:injection, type:test	3	Mon
AGD-069	Eval harness: precision/recall/F1 (per-release gate)	P0	module:injection, type:test	3	Tue
AGD-071	OWASP LLM Top 10 coverage map: ATTACK_MATRIX.md	P0	module:injection, type:docs	2	Wed
AGD-072	PyPI v1.0.0 release via GitHub Actions release workflow	P0	module:infra, type:feature	3	Thu-Fri

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ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-073	Release notes + CHANGELOG.md + GitHub Release	P1	type:docs	2	Fri

■ v1.0 GATE — End of Week 8 | Production-grade on PyPI | Actively interview now

PHASE 3 — OPTION A: PILOT READY (Sprints 9–12 · Weeks 9–12)

Sprint 9 — SSO/RBAC

Week / Phase	9 · Option A Phase 2
Sprint Goal	Enterprise SSO via SAML 2.0 / OIDC · RBAC policies enforced at API layer · Okta / Azure AD integration
Demo Moment	Okta login working — enterprise prospect can authenticate

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-074	SAML 2.0 SP implementation (python3-saml)	P0	module:api, type:feature	5	Mon-Tue
AGD-075	OIDC / OAuth 2.0 integration (authlib)	P0	module:api, type:feature	5	Tue-Wed
AGD-077	RBAC: row-level security for multi-tenant event filtering	P0	module:api, type:feature	5	Thu-Fri

Sprint 10 — Kubernetes / Helm Chart + Docker Packaging

Week / Phase	10 · Option A Phase 2
Sprint Goal	Docker image published to GHCR · Helm chart for K8s deployment · HPA configured
Demo Moment	Helm install agentguard deploys full stack to kind cluster in under 5 minutes

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-079	Multi-stage Dockerfile (<200MB final image)	P0	module:infra, type:feature	3	Mon
AGD-081	Helm chart: api + worker + dashboard deployments	P0	module:infra, type:feature	5	Tue-Thu
AGD-082	HPA: autoscale worker on redis queue depth metric	P1	module:infra, type:feature	3	Thu-Fri
AGD-084	K8s smoke tests: deploy to kind, run scenario tests	P0	module:infra, type:test	3	Fri

Sprint 11 — Next.js Dashboard v2

Week / Phase	11 · Option A Phase 2
Sprint Goal	Replace Streamlit with Next.js 14 + Tailwind + shadcn/ui · WebSocket real-time · RBAC role-gated views
Demo Moment	Enterprise dashboard with live WebSocket feed and role-gated views for demo to pilot prospect

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ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-085	Next.js 14 app scaffold with Tailwind + shadcn/ui	P0	module:dashboard, type:feature	3	Mon
AGD-086	WebSocket real-time event feed	P0	module:dashboard, type:feature	5	Tue-Wed
AGD-087	4 role-gated views (Admin/SOC/Compliance/Viewer)	P0	module:dashboard, type:feature	5	Wed-Thu
AGD-089	One-click PDF compliance report (SOC2 + HIPAA mapping)	P0	module:dashboard, type:feature	5	Thu-Fri

Sprint 12 — Pilot Prep + Compliance Templates + v1.5 Release

Week / Phase	12 · Option A Phase 2
Sprint Goal	SOC2 CC6 + HIPAA §164.312 compliance report templates · Pilot onboarding guide · v1.5.0 release
Demo Moment	■ Pilot customer can receive SOC2 and HIPAA audit evidence pack from AgentGuard in one click

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-091	SOC2 CC6 compliance report template (auto-populated)	P0	module:audit, type:feature	5	Mon-Tue
AGD-092	HIPAA §164.312 audit evidence pack template	P0	module:audit, type:feature	5	Tue-Wed
AGD-094	Pilot customer onboarding guide: 5-min setup walkthrough	P0	type:docs	3	Thu
AGD-096	v1.5.0 PyPI release + release notes	P0	module:infra, type:feature	2	Fri

■ v1.5 GATE — End of Week 12 | Pilot-Ready | SSO + K8s + Next.js + Compliance Reports

PHASE 4 — OPTION A: PRODUCTION SCALE (Sprints 13–16 · Weeks 13–16)

Sprint 13 — Multi-Agent Federation + Cross-System Session Tracking

Week / Phase	13 · Option A Phase 3
Sprint Goal	Track agent interactions across distributed multi-agent systems · Cross-session correlation · Federation governance
Demo Moment	Agent call graph visualization showing cross-system session timeline

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-097	Distributed session ID propagation (W3C TraceContext-inspired)	P0	module:interceptor, type:feature	5	Mon-Wed
AGD-098	Cross-agent call graph visualization in dashboard	P1	module:dashboard, type:feature	5	Thu-Fri
AGD-099	Multi-deployment governance: single instance, multiple clusters	P0	module:api, type:feature	5	Wed-Fri

Sprint 14 — Policy Marketplace + PyPI Public Launch

Week / Phase	14 · Option A Phase 3
Sprint Goal	Community policy templates marketplace · pip install agentguard[finance] bundles industry rules · v2.0-alpha
Demo Moment	■ pip install agentguard[healthcare] auto-installs HIPAA policy templates

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-101	Policy marketplace: community YAML templates (4 verticals)	P1	module:policy, type:feature	5	Mon-Thu
AGD-102	Optional extras: agentguard[finance/healthcare/legal]	P1	module:infra, type:feature	3	Thu
AGD-103	v2.0-alpha PyPI release + Hacker News Show HN	P0	module:infra, type:feature	2	Fri

Sprint 15 — S3 Audit Archival + Multi-Tenant SaaS Foundation

Week / Phase	15 · Option A Phase 3
Sprint Goal	Long-term audit log archival to S3/GCS · Multi-tenant org isolation · SaaS groundwork

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Demo Moment	Multi-tenant demo: two orgs with completely isolated event streams

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-105	S3/GCS archival: nightly export to immutable storage	P0	module:audit, type:feature	5	Mon-Tue
AGD-106	Multi-tenant: org-level isolation (schema-per-tenant)	P0	module:infra, type:feature	8	Wed-Fri
AGD-107	Tenant onboarding API: create org, assign admin, provision	P1	module:api, type:feature	5	Thu-Fri

Sprint 16 — Performance Hardening + v2.0 Release

Week / Phase	16 · Option A Phase 3
Sprint Goal	Full performance validation at production scale · v2.0.0 public release
Demo Moment	■ v2.0-ALPHA publicly released on PyPI with blog post on stridemap.ai

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-109	Nightly Locust: 500 events/sec sustained (60s)	P0	module:infra, type:test	3	Mon
AGD-110	Latency p99 validation <50ms overhead under load (weekly gate)	P0	module:infra, type:test	3	Tue
AGD-112	Anthropic + OpenAI Agents SDK native support	P1	module:interceptor, type:feature	5	Wed-Thu
AGD-111	v2.0.0 release notes + migration guide from v1.5	P0	type:docs	3	Fri

■ v2.0 ALPHA — End of Week 16 | Multi-tenant · S3 · 500 events/sec · Public OSS

PHASE 5 — OPTION C: AGENT RUNTIME SECURITY (Sprints 17–24 · Weeks 17–24)

Goal: Evolve from governance (observe + policy) to runtime security (OS-level isolation). No other open-source AI governance tool does OS-level sandboxing — this is the long-term technical moat.

Sprints 17–18 — seccomp/eBPF Research + Prototype

Weeks / Phase	17–18 · Option C
Sprint Goal	eBPF probe intercepting syscalls from agent tool processes · seccomp profile generation
Demo Moment	eBPF probe demo: file open and network connect syscalls from agent PID captured in real time

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-113	eBPF probe: intercept file open + network connect from agent PIDs	P0	module:infra, type:feature	8	Week 17
AGD-114	seccomp profile generator: auto-build allowlist from tool dry-run	P1	module:infra, type:feature	5	Week 17
AGD-116	ADR: gVisor vs eBPF vs ptrace for Option C	P0	type:docs	2	Week 18

Sprints 19–20 — Tool Call Process Isolation

Weeks / Phase	19–20 · Option C
Sprint Goal	Each tool call runs in isolated subprocess · Filesystem isolation via bind mounts · Network namespace per invocation
Demo Moment	Isolation demo: tool attempting filesystem write outside bind mount is silently denied

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-117	Process isolation wrapper: tool in subprocess with seccomp profile	P0	module:infra, type:feature	8	Week 19
AGD-118	Filesystem isolation: read-only bind mount for tool process	P0	module:infra, type:feature	5	Week 19
AGD-120	Performance benchmark: isolation overhead <100ms per invocation	P0	module:infra, type:test	3	Week 20

Sprints 21–22 — Network Egress Control + Process Spawn Prevention

Weeks / Phase	21–22 · Option C
Sprint Goal	Kernel-level network egress enforcement · Block process spawning from agent tool code

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Demo Moment	Egress block demo: agent tool attempting to connect to non-allowlisted IP is blocked at kernel level

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-121	eBPF XDP network filter: block egress to non-allowlisted IPs	P0	module:infra, type:feature	8	Week 21
AGD-122	Process spawn prevention: SECCOMP_RET_KILL on execve from agents	P0	module:infra, type:feature	5	Week 22
AGD-123	Lateral movement detection: flag agent accessing out-of-scope resources	P0	module:interceptor, type:feature	5	Week 22

Sprints 23–24 — Option C Integration + v2.0 Final Release

Weeks / Phase	23–24 · Option C
Sprint Goal	Full Option C integration with governance layer · v2.0.0 final release with runtime security docs
Demo Moment	■■ v2.0 FINAL: complete governance + runtime security. Conference talk proposal submitted.

ID	Title	Priority	Labels	Pts	Day/Sprint
AGD-124	Integrate Option C runtime isolation with Option A governance	P0	module:infra, type:feature	8	Week 23
AGD-125	Option C scenario tests: escape, lateral movement, spawn prevention	P0	module:infra, type:test	5	Week 23
AGD-126	v2.0.0 final release: changelog + Option C setup guide	P0	type:docs	3	Week 24
AGD-127	Conference talk proposal: NGFW mental models for AI agents	P2	type:demo	2	Week 24

■■ v2.0 FINAL — End of Week 24 | Full Runtime Security | Enterprise-Grade | Complete

Appendix A — Master Sprint Summary

Sprint	Week	Phase	Goal	Key Deliverable	Demo Moment
1	1	MVP	Foundation	stridemap.ai live, CI green	Website live
2	2	MVP	Interceptor + Dashboard	SQLite event store, Streamlit	Live trace viewer
3	3	MVP	Policy + DLP	YAML rules, Presidio PII	BLOCK demo
4	4	MVP	Injection + Audit	3 attack scenarios	■ BIG DEMO
5	5	Opt-A P1	FastAPI	10 REST endpoints, JWT	Postman collection
6	6	Opt-A P1	PostgreSQL + Redis	100+ events/sec	Locust benchmark
7	7	Opt-A P1	SIEM + Prometheus	CEF webhooks	Splunk events
8	8	Opt-A P1	Classifier v2 + PyPI	v1.0.0 on PyPI	■ v1.0 RELEASE
9	9	Opt-A P2	SSO/RBAC	SAML + OIDC	Okta login demo
10	10	Opt-A P2	Kubernetes	Helm chart	K8s deploy in 5 min
11	11	Opt-A P2	Next.js Dashboard	WebSocket, RBAC views	Enterprise UI
12	12	Opt-A P2	Pilot Prep	SOC2/HIPAA templates	■ v1.5 PILOT READY
13	13	Opt-A P3	Multi-agent	Cross-system sessions	Agent graph viz
14	14	Opt-A P3	Policy Marketplace	Vertical rule packs	Community templates
15	15	Opt-A P3	S3 + Multi-tenant	Immutable archival	Tenant isolation
16	16	Opt-A P3	Performance	500 events/sec	■ v2.0-ALPHA
17-18	17-18	Opt-C	eBPF Research	Syscall interception	eBPF probe demo
19-20	19-20	Opt-C	Process Isolation	Namespaced tool calls	Isolation demo
21-22	21-22	Opt-C	Network Egress	Kernel-level filtering	Egress block demo
23-24	23-24	Opt-C	Integration	Full runtime security	■■ v2.0 FINAL

Appendix B — Job Search Interview Story Map

Job Role	Key Skills	AgentGuard Evidence	Sprint(s)
AI Security Engineer	Threat modeling, LLM security, OWASP	OWASP LLM Top 10 coverage, injection detection, DLP scanner	S3, S4, S7
Platform / Infra Engineer	Kubernetes, Docker, CI/CD, Python async	Helm chart, GitHub Actions, monorepo, Redis queue	S1, S6, S10
Backend Engineer (Python)	FastAPI, PostgreSQL, async, REST API	FastAPI + asyncpg + Redis + JWT + RBAC	S5, S6
Product Security / AppSec	Policy-as-code, SIEM, SOC2/HIPAA	YAML policies, CEF format, SOC2/HIPAA templates	S3, S7, S12

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Job Role	Key Skills	AgentGuard Evidence	Sprint(s)
Staff / Principal Engineer	Architecture, ADRs, system design	8-layer governance stack, ADRs doc, fail-open design decisions	All
Developer Relations	OSS, documentation, demos, community	PyPI package, Loom videos, README quality, marketplace	S4, S8, S14

Appendix C — Learning Resources Per Module

Module	Primary Sources	Interview Depth Target
LangGraph Interceptor	LangGraph BaseCallbackHandler source, LangSmith tracing docs	Explain callback lifecycle, run_id propagation, async safety
Policy Engine	Palo Alto NGFW policy docs, NIST SP 800-207 (Zero Trust)	Design policy schema from scratch, explain priority ordering
DLP Scanner	Microsoft Presidio docs, GDPR Article 4, HIPAA Safe Harbor	Entity recognition vs classification, local vs cloud tradeoff
Injection Detection	OWASP LLM Top 10, Perez et al. "Ignore Previous Prompt" paper	Direct vs indirect injection, two-stage architecture rationale
Audit Chain	NIST SP 800-92 (log management), SHA-256 spec	Cryptographic chaining, non-repudiation for compliance
FastAPI + JWT	FastAPI docs, RFC 7519 (JWT), OWASP REST Security Cheat Sheet	Implement JWT auth, explain expiry/refresh tradeoffs
Kubernetes	K8s docs, Helm best practices, HPA custom metrics	AgentGuard pod topology, HPA trigger metric design

Appendix D — Companion Files

AgentGuard_Sprint_Plan_v1.md — Notion-ready markdown — import directly into Notion workspace

AgentGuard_Linear_Import.csv — Linear import file — 96 tickets with titles, descriptions, priorities, labels, and cycles

AgentGuard_PRD_v1.pdf — Product Requirements Document — full module requirements, personas, acceptance criteria

AgentGuard_Roadmap_v1.pdf — Product Roadmap — 24-sprint plan with daily breakdowns for Sprints 1–4

AgentGuard_System_Architecture_v1.pdf — System Architecture — 8-layer architecture, component deep-dives, ADRs

AgentGuard_System_Design_v1.pdf — System Design — data models, API contracts, storage evolution

AgentGuard_Functional_Spec_v1.pdf — Functional Specification — complete behavioural contract for all 5 modules

AgentGuard_Test_Plan_v1.pdf — Test Plan — 5-layer test framework, evaluation metrics, CI/CD gates

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